Computational Complexity

Reading guide and Homework for **Week 06**

1 Reading Guide

You can read about:

- L, NL, reachability is NL-complete under logspace-reductions. §4.3.
- You should study the proof that $\mathsf{NL} = \mathsf{coNL}$, also in §4.3, though we didn't have enough time to cover it in class.
- We won't cover this in the course, but there is a remarkable proof that connectivity in undirected graphs is in L. §21.4.
- The Baker-Gill-Solovay Theorem (the Relativization Barrier). §3.4.

There were several details that we covered very quickly, and which **must** be looked at carefully, in order to properly understand what is going on. This kind of familiarity will be necessary to understand what's being said in class.

2 Homework

Skill: *Reason about space.*

1. Exercise 4.7 of the book.
2. Exercise 4.8.
3. (Exercise 4.5 of the book) Show that 2SAT is in NL. For this consider the following hints:

   Hint 1. Suppose your formula has a clause $x_1 \vee x_2$. Note that any satisfying assignment with $x_1 = 0$ must have $x_2 = 1$. The NL-complete problem is directed reachability. Can you create a graph where reachability means something.

   Hint 2. Create a graph with $2n$ nodes, one for each equality $x_i = 0/1$. Add the above implications (e.g. for clause $x_1 \vee x_2$, one would add the two directed edges $x_1 = 0 \to x_2 = 1$ and $x_2 = 0 \to x_1 = 1$). What does this graph tell you?

4. Prove the *space hierarchy theorem*, that if $f(n) = o(g(n))$, then $\mathsf{DSPACE}(g) \not\subseteq \mathsf{DSPACE}(f)$. (Observe that it is tighter than the time hierarchy theorem, i.e., we do not need the log factor.)
5. Consider the problem *product of permutations*, as follows. We are given as input $m + 1$ permutations $\sigma_1, \ldots, \sigma_m, \sigma$ of $[n]$. We wish to know if the product $\sigma_1 \cdot \cdots \cdot \sigma_m$ equals σ.

   Convince yourself that there is a reasonable way of encoding a permutation of $[n]$ using $\Theta(n \log n)$ bits.

   Show that the problem can be solved in L.

Skill: *Show completeness under logspace reductions.*

6. Exercise 4.9.

7. (Exercise 4.4 of the book) A directed graph is called *strongly connected* if there exists a (directed) path between any two vertices. Show that the problem of deciding whether a given directed graph is strongly connected is NL-complete under logspace reductions.

Skill: *Reason about the BGS theorem.*

8. Show that the time hierarchy theorem relativizes. It follows, for example, $\mathsf{P}^A \neq \mathsf{E}^A$ for every oracle A.

9. Show that the non-deterministic time-hierarchy theorem relativizes.

10. Exercises 3.7 and 3.8.

Skill: *Learn new concepts by yourself. (The test will have questions about these, possibly asking you to prove something that appears in these sections, or reason in a way that requires that you understood them.).*

11. Learn the proof that $\mathsf{NL} = \mathsf{coNL}$ (§4.3).